PSTAT 10 Data Science Principles

Lecture 11: Continuous Random Variables

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August 20, 2025

Introduction

Review: Lecture 10

👈 Last lecture we had a very math heavy lecture covering:

- ▶️ Random Variables (RVs)
- ▶️ Probability Mass Functions (PMFs)
- ▶️ Cumulative Distribution Functions (CDFs)
- ▶️ Binomial Distribution
- ▶️ `rbinom`, `pbinom` and `dbinom` functions

Overview: Lecture 11

👇 This lecture is again a little math heavy and we shall be looking at:

- ▶️ Generalizing `rbinom`, `pbinom` and `dbinom` functions
- ▶️ Continuous RVs
- ▶️ Continuous Uniform Distribution
- ▶️ Normal (Gaussian) Distribution

Recap

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Recall the following definitions:

- A random variable X is a mathematical object taking numerical outputs corresponding to a random experiment.

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- A probability mass function (PMF) of random variable X is defined by

$$p_X(k) = \mathbb{P}(X = k),$$

  which is non-zero for values k in the support of X.
- The cumulative distribution function (CDF) of a random variable X is defined by

$$F_X(k) = \mathbb{P}(X \leq k).$$

Recap - Binomial Distribution

Consider a random variable X that follows a Binomial (n, p) distribution.

The PMF and CDF of X respectively are

$$p_X(k) = \mathbb{P}(X = k) = \binom{n}{k} \cdot p^k \cdot (1 - p)^{n-k}$$

$$F_X(k) = \mathbb{P}(X \leq k) = \mathbb{P}(X = 0) + \mathbb{P}(X = 1) + \ldots + \mathbb{P}(X = k)$$

We have a quick practice with the `rbinom`, `pbinom` and `dbinom` functions.

Recap - `dbinom()` Function

To determine the probability of X taking specific values (i.e. $\mathbb{P}(X = k)$) we use the `dbinom()` function.

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Examples

1. *What is* $\mathbb{P}(X = 7)$ *for* $X \sim \text{Binomial}\ (n = 15, p = 0.69)$*?*

```r
dbinom(x = 7, size = 15, prob = 0.69)
```

```
[1] 0.04086822
```

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```

```
[1] 0.04086822
```

2. *What is* $\mathbb{P}(Y = 1)$ *for* $Y \sim \text{Binom}\ (n = 4, p = 0.24)$*?*

```r
dbinom(x = 1, size = 4, prob = 0.25)
```

```
[1] 0.421875
```

Recap - `pbinom()` Function

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Examples

1. *What is* $\mathbb{P}(X \leq 7)$ *for* $X \sim \text{Binomial}\ (n = 11, p = 0.51)$*?*

```r
pbinom(q = 7, size = 11, prob = 0.51)
```

```
[1] 0.8733076
```

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Examples

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```
pbinom(q = 7, size = 11, prob = 0.51)
```

```
[1] 0.8733076
```

2. *What is* $\mathbb{P}(X \leq 4)$ *for* $X \sim \text{Binomial}\ (n = 5, p = 0.91)$*?*

```
pbinom(q = 4, size = 5, prob = 0.91)
```

```
[1] 0.3759679
```

Recap - `rbinom()` Function

To generate realizations from a $X \sim \text{Binomial}\ (n, p)$ RV we use the `rbinom()` function.

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Examples

1. *Generate 10 realizations from* $X \sim \text{Binomial}\ (20, 0.6)$ *RV.*

```r
rbinom(n = 10, size = 20, prob = 0.6)
```

```
 [1] 12 14 14 10 12 14 13 12 14 10
```

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Examples

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```r
rbinom(n = 10, size = 20, prob = 0.6)
```

```
 [1] 12 14 14 10 12 14 13 12 14 10
```

2. *Use 1000 realizations to approximate the expected value of* $X \sim \text{Binomial}\ (30, 0.45)$.

```r
rbinom(n = 1000, size = 30, prob = 0.45) |>
    mean()
```

```
[1] 13.604
```

Poisson Distribution

General Functions

These functions each take the general form of the base R functions that exist for the key named distributions (some of which we will cover later).

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We summarize these functions in the table below:

Table: R Base Discrete Distribution Functions

Binomial	Poisson	Geometric
`dbinom()`	`dpois()`	`dgeom()`
`pbinom()`	`ppois()`	`pgeom()`
`qbinom()`	`qpois()`	`qgeom()`
`rbinom()`	`rpois()`	`rgeom()`

Table 2: R Base Discrete Distribution Functions.

Note that the discrete uniform is different since we use the `sample()` *function instead.*

Poisson Distribution

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A Poisson random variable describes the outcomes of an experiment counting occurrences in a fixed time interval when the mean rate of occurrence is known.

Formally, the PMF of the Poisson random variable $X \sim \mathcal{P}(\lambda)$ with mean rate λ is:

$$\mathbb{P}(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}; \quad k \in \{0, 1, 2, ...\}.$$

The CDF of this distribution is quite complicated mathematically and is not covered in this course.

Poisson Distribution Plot

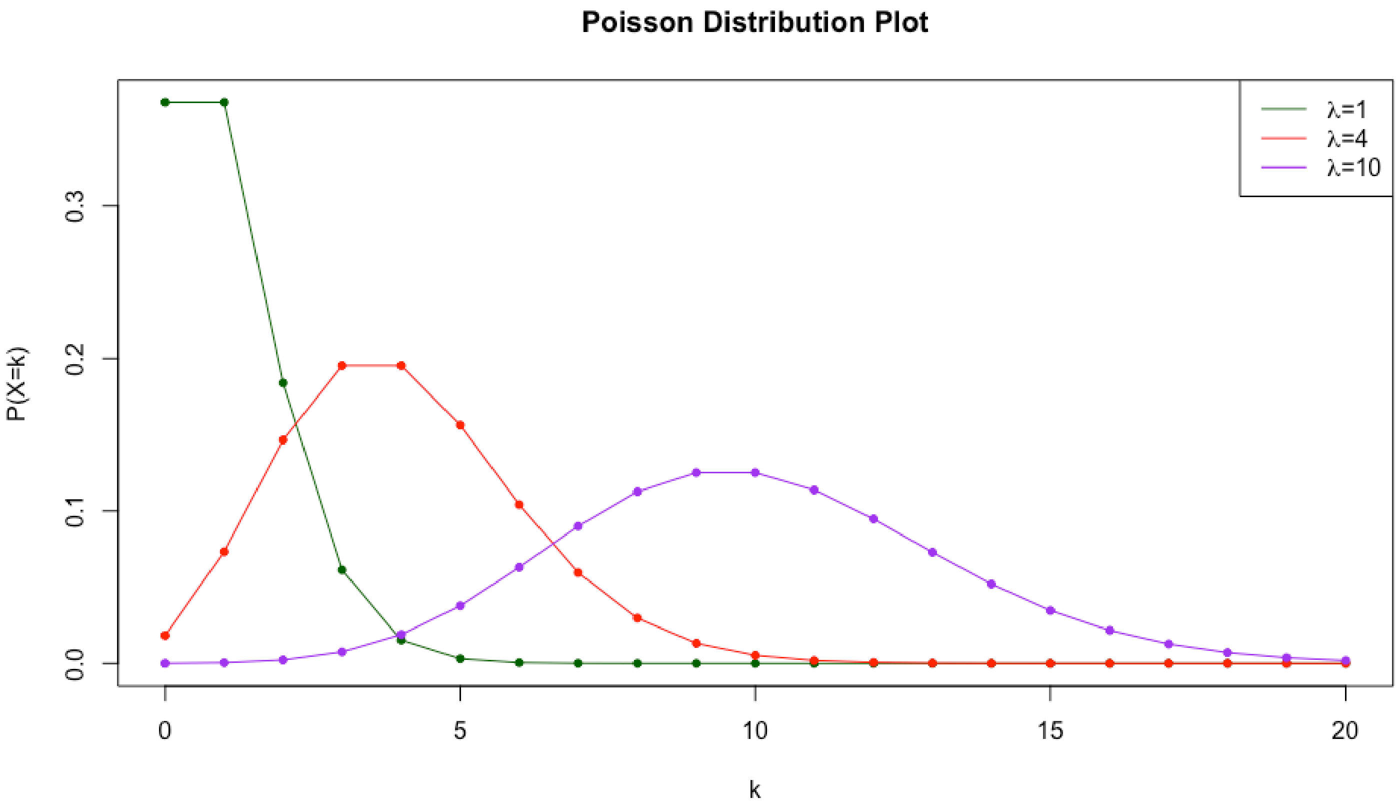

Figure 1: Poisson distribution plot

Poisson Distribution Functions

Recall the roles of the functions `dpois()`, `ppois()` and `rpois()`. Below we demonstrate how to use them.

Examples

1. *Compute* $\mathbb{P}(X = 3)$ *when* $X \sim \mathcal{P}(2)$.

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```r
dpois(3, 2)
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[1] 0.180447
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2. *Compute* $\mathbb{P}(X > 5)$ *when* $X \sim \mathcal{P}(3)$.

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1. *Compute* $\mathbb{P}(X = 3)$ *when* $X \sim \mathcal{P}(2)$.

```r
dpois(3, 2)
```

```
[1] 0.180447
```

2. *Compute* $\mathbb{P}(X > 5)$ *when* $X \sim \mathcal{P}(3)$.

```r
1 - ppois(4, 3)
```

```
[1] 0.1847368
```

Poisson Distribution Functions

3. *Using 10000 simulations, estimate the mean of* X *when* $X \sim \mathcal{P}(5)$.

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```r
rpois(10000, 3) |> mean()
```

```
[1] 3.0021
```

Continuous Random Variables

Motivation

So far we have been considering discrete random variables i.e. random variables with discrete support:

- Bernoulli $\{0, 1\}$
- Binomial $\{0, 1, \ldots, n\}$
- Discrete Uniform $\{\text{item\_1}, \text{item\_2}, \ldots, \text{item\_k}\}$
- Anything countable $\{\ldots, -2, -1, 0, 1, 2, \ldots\}$

What about continuous random variables with continuous support:

- Closed intervals $[0, 1]$
- Open or partially open intervals $(0, 1)$ and $[0, 1)$
- Times, heights, weights, position on a dartboard or in a cube etc.

Probability Paradox

Remember that $\mathbb{P}(X = k) \neq 0$ for any value k in the support of a discrete random variable X.

This is not true for a continuous random variable!

- Recall we discussed that the probability of selecting a single number from a continuous interval is zero.
- However, the probability for selecting a subinterval from a continuous interval is nonzero.

Probability Density Functions

- Discrete random variables are described by mass functions.
  - → Probabilities are point masses so it makes sense to discuss $\mathbb{P}(X = k)$.
- Continuous random variables are described by density functions.
  - → Probabilities are no longer point masses but are instead described by areas under distribution functions.
  - → This makes intuitive sense as to why $\mathbb{P}(X = k)$ no longer makes sense as the area under a single point of a curve is 0.

Continuous Uniform Distribution

Motivating Example

To consider a practical example, suppose buses arrive at a stop every 12 minutes. If you arrive at the bus stop at a random time, how long will you wait?

- ▶️ In the best case scenario?
- ▶️ In the worst case scenario?

Motivating Example

To consider a practical example, suppose buses arrive at a stop every 12 minutes. If you arrive at the bus stop at a random time, how long will you wait?

- ▶️ In the best case scenario?
- ▶️ In the worst case scenario?

Let X be the time you wait for the bus, i.e. a continuous random variable supported on $(0, 12)$.

To select a distribution we must consider what assumptions to make.

Distribution Assumptions

To make a claim about the distribution of X we make the following assumptions:

- We arrive at the bus stop at a random time and
- You are equally likely to arrive at any point between the n-th and $(n+1)$-st bus.

This is the continuous version of the uniform distribution hence we define a continuous uniform random variable.

Note again, don't worry too much about the math. Focus on the intuition!

Continuous Uniform Distribution

A continuous uniform random variable $X \sim \mathcal{U}([a,b])$ with continuous support $[a,b]$ has probability density function

$$f_X(x) = \begin{cases} \frac{1}{b-a} & x \in [a,b] \\ 0 & \text{otherwise} \end{cases}.$$

In the case of our bus stop example the support is $[0,12]$ i.e. $a = 0$ and $b = 12$ and so for $X \sim \mathcal{U}([0,12])$ we have that

$$f_X(x) = \frac{1}{12}; \quad x \in [0,12].$$

Continuous Uniform Distribution

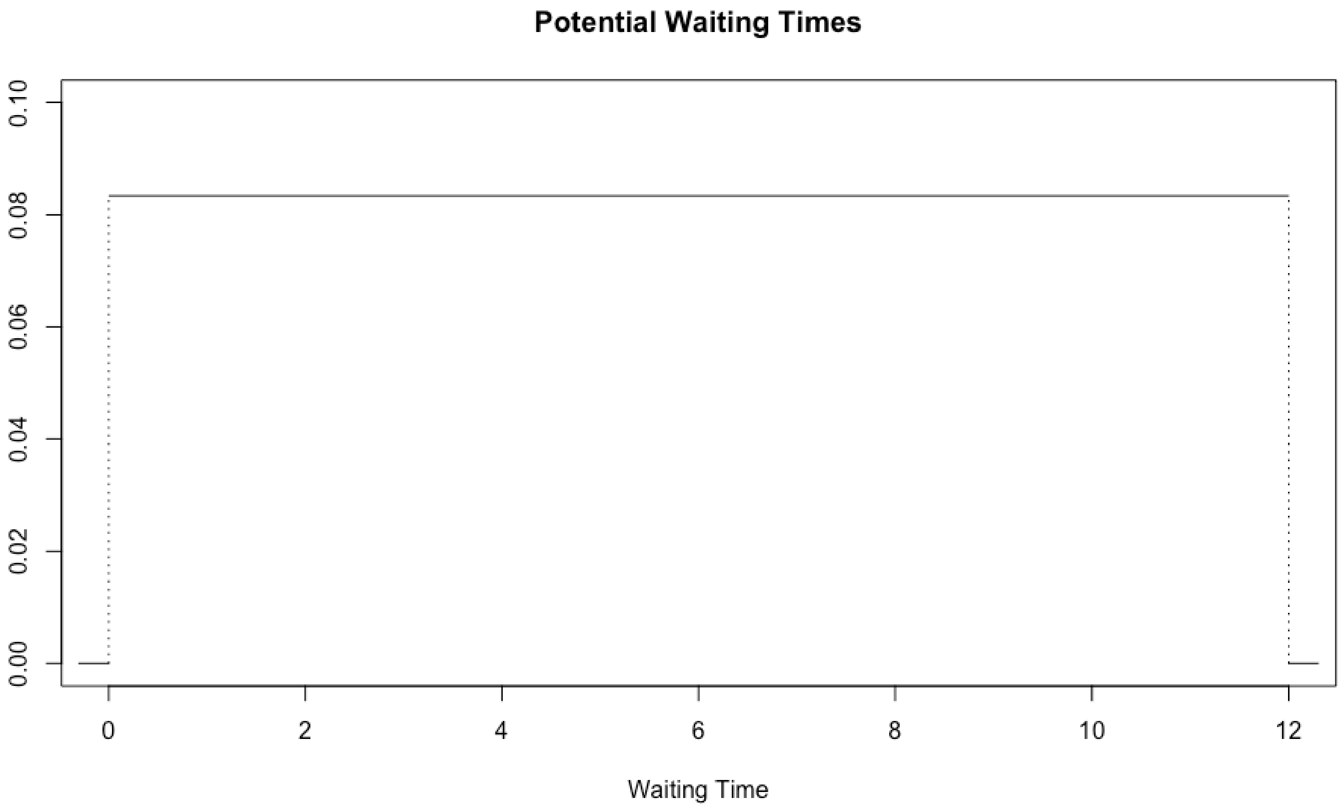

Figure 2: Probability Density Plot

Continuous Uniform CDF

You may have noticed that since we can only compute the probability that X falls in an interval then we are primarily interested in the CDF of continuous random variables since

$$\mathbb{P}(p \leq X \leq q) = \mathbb{P}(X \leq q) - \mathbb{P}(X \leq p) = F_X(q) - F_X(p).$$

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Often it is challenging to write a closed form expression for the CDF but in the case of the continuous uniform we find that the CDF is given by:

$$F_X(x) = \mathbb{P}(X \leq x) = \frac{x-a}{b-a}; \quad x \in [a, b].$$

Uniform Probabilities

Lets say we wish to compute analytically the probability that the wait time is between 3 and 7 minutes. This is equivalent to computing the shaded area

Uniform Probabilities

Figure 3: Probability Density Plot

Uniform Integral

This can clearly be computed using basic geometry but to practice our calculus methods we have that

$$\begin{aligned}\mathbb{P}(3 \leq X \leq 7) &= \int_3^7 f_X(x)dx \\ &= \int_3^7 \frac{1}{12}dx \\ &= \left[\frac{x}{12}\right]_3^7 \\ &= \frac{7}{12} - \frac{3}{12} = \frac{4}{12} = \frac{1}{3}.\end{aligned}$$

Uniform CDF Computation

We compare this to the result when we use the CDF:

$$\begin{aligned}\mathbb{P}(3 \leq X \leq 7) &= \mathbb{P}(X \leq 7) - \mathbb{P}(X \leq 3) \\ &= \frac{7-0}{12-0} - \frac{3-0}{12-0} = \frac{1}{3}.\end{aligned}$$

Uniform CDF Computation

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In fact using calculus, we can derive the CDF by computing:

$$\begin{aligned}F_X(x) = \mathbb{P}(X \leq x) &= \int_a^x f_X(x)dx \\ &= \int_a^x \frac{1}{b-a}dy = \left[\frac{y}{b-a}\right]_a^x \\ &= \frac{x-a}{b-a}.\end{aligned}$$

`punif` Function

R has the in-built function `punif()` for computing the uniform CDF $\mathbb{P}(X \leq x)$ which has the syntax:

```
punif(q = x, min = a, max = b, lower.tail = TRUE)
```

- `q` is the quantile x
- `min` is the lower bound of the support a
- `max` is the upper bound of the support b.

Checking out computations earlier we have that:

```
punif(7, 0, 12) - punif(3,0,12)
```

```
[1] 0.3333333
```

Continuous Uniform R Functions

Much like with discrete random variables we have 4 standard functions for each distribution:

Uniform	Exponential	Normal	Role
`dunif()`	`dexp()`	`dnorm()`	PDF
`punif()`	`pexp()`	`pnorm()`	CDF
`qunif()`	`qexp()`	`qnorm()`	Quantile
`runif()`	`rexp()`	`rnorm()`	Simulation

Exercise - Continuous Uniform in R

For the next 4 minutes, use the functions from the previous slide to answer the following questions about the continuous uniform random variable $X \sim \mathcal{U}(10, 25)$:

1. *What is the probability that X takes a value greater than 20? (Hint: You can set lower.tail = FALSE)*
2. *What is the probability that X takes a value either greater than 19 or less than 15?*
3. *Use simulation to approximate the expected value of X.*

Solution - Continuous Uniform in R

```r
punif(20, min = 10, max = 25, lower.tail = FALSE)
punif(19, min = 10, max = 25, lower.tail = FALSE) +
  punif(15, min = 10, max = 25, lower.tail = TRUE)
runif(10000, min = 10, max = 25) |>
    mean()
```

```
[1] 0.3333333
```

```
[1] 0.7333333
```

```
[1] 17.55003
```

Normal Distribution

What is the Normal Distribution

The normal (Gaussian) distribution is the most important distributions in statistics. It is more commonly known as the bell curve.

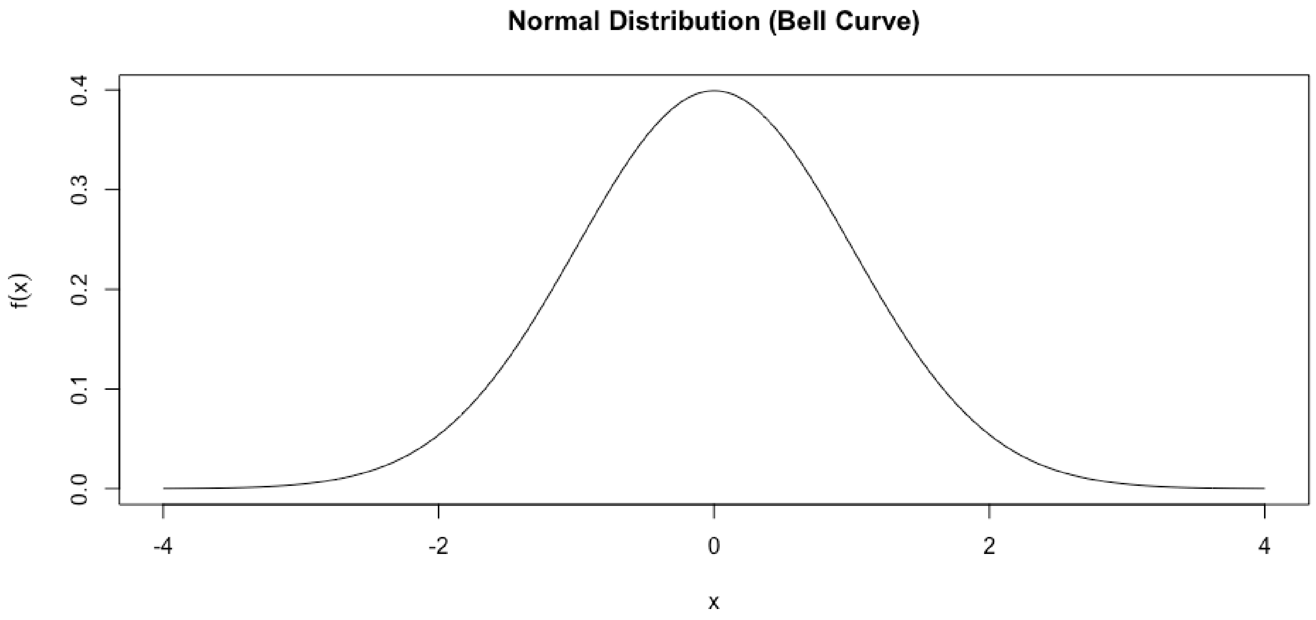

Figure 4: Distribution plot for a standard normal random variable.

Normal Parameters

The normal distribution is described by parameters:

- The mean μ describes the location of the peak; and
- The variance σ^2 describes the spread of the bell.

We denote a normal random variable $X \sim \mathcal{N}(\mu, \sigma^2)$.

In our previous plot above we used $\mu = 0$ and $\sigma^2 = \sigma = 1$ which is known as a standard normal distribution.

Normal Parameter Plots

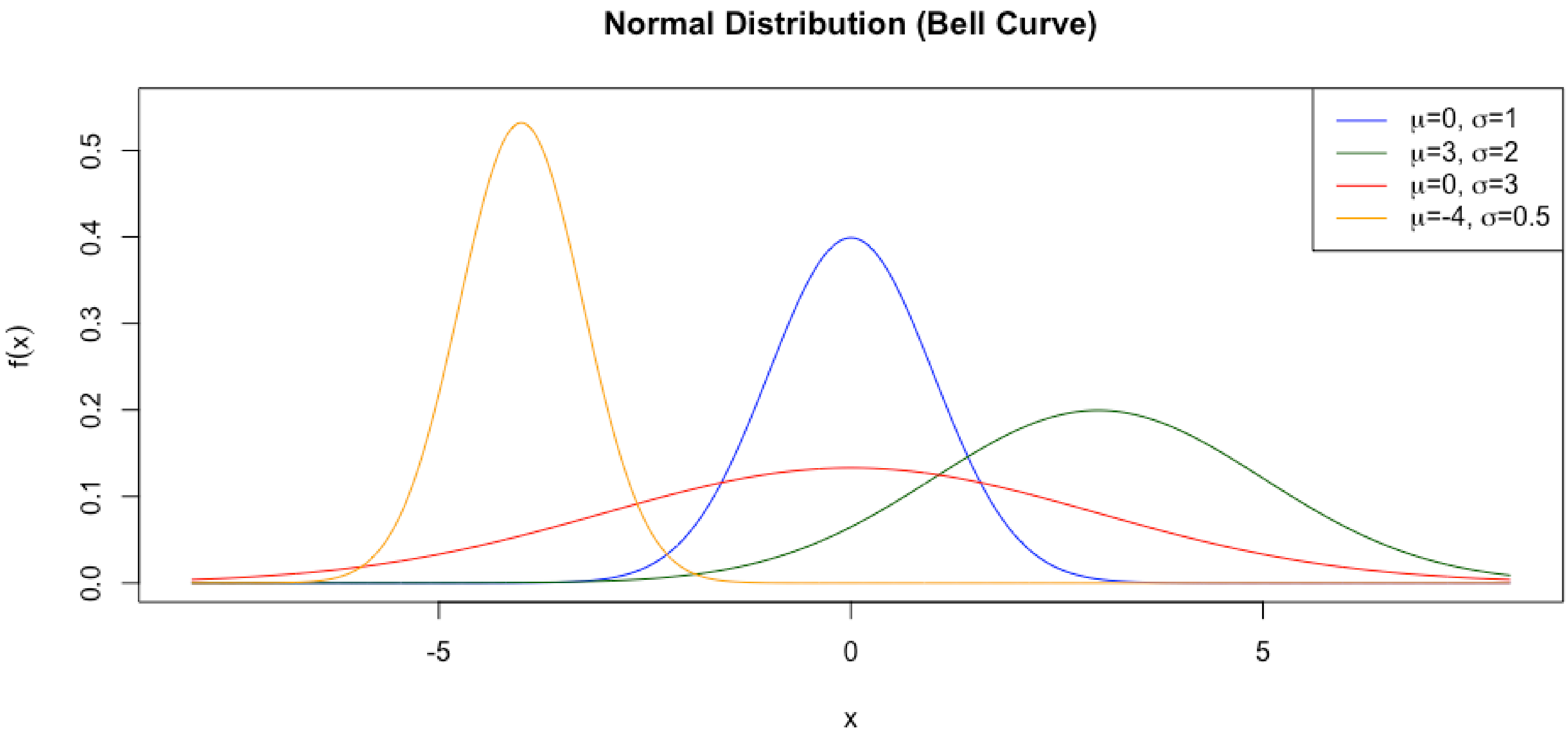

Figure 5: Multiple Normal Distributions

Normal Distribution Function

For normal random variable $X \sim \mathcal{N}(\mu, \sigma^2)$ the density function is

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{-(x-\mu)^2}{2\sigma^2}\right); \quad x \in \mathbb{R}.$$

Hence if we wish to determine $\mathbb{P}(-1 \leq X \leq 1)$ we determine

$$\mathbb{P}(-1 \leq X \leq 1) = \int_{-1}^{1} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{-(x-\mu)^2}{2\sigma^2}\right) dx,$$

which isn't so easy to compute. Instead we can use R functions!

Examples - Normal R Functions

1. Compute $\mathbb{P}(3 \leq X \leq 10)$ when $X \sim \mathcal{N}(4, 3^2)$.

```r
pnorm(10, mean = 4, sd = 3, lower.tail = TRUE) -
    pnorm(3, mean = 4, sd = 3, lower.tail = TRUE)
```

```
[1] 0.6078085
```

2. Approximate the mean of the sum of two $X \sim \mathcal{N}(1, 2^2)$ random variables using $n = 10000$ simulations.

```r
(rnorm(10000, mean = 1, sd = 2) + rnorm(10000, mean = 1, sd = 2))  |>
    mean()
```

```
[1] 2.02962
```

Exercise - Normal Probability

Suppose the heights of students in class A are normally distributed with a mean $\mu = 67$ inches and a standard deviation of $\sigma = 3$ inches?

1. *What is the probability that a randomly selected students height is between 63 and 70 inches?*

Now consider class B student heights which we suppose are normally distributed with mean $\mu = 70$ inches and a standard deviation of $\sigma = 4$ inches.

2. *Use* $n = 10000$ *simulations to determine the probability that a randomly selected student from class A is taller than a randomly selected student from class B assuming the class heights are independent.*

Solution - Normal Probability

```r
pnorm(70, mean = 67, sd = 3) - pnorm(63, mean = 67, sd = 4)
```

```
[1] 0.6826895
```

```r
(rnorm(10000, mean = 67, sd = 3) > rnorm(10000, mean = 70, sd = 4)) |> mean()
```

```
[1] 0.2828
```

Interesting Result

Stepping outside the bounds of the course for a moment we take a look at this last simulation.

Define $X \sim \mathcal{N}(67, 3^2)$ and $Y \sim \mathcal{N}(70, 4^2)$.

We computed:

$$\mathbb{P}(X > Y) = \mathbb{P}(X - Y > 0)$$

Using probability theory results we can show that $X - Y \sim \mathcal{N}(-3, 5^2)$ and so we can compute this probability explicitly as:

```
pnorm(0, mean = -3, sd = 5, lower.tail = F)
```

```
[1] 0.2742531
```

Summary

Topics Covered

🤔 Today we covered a lot of material including:

- ▶️ Poisson Distributions
- ▶️ Generalizing R Functions
- ▶️ Continuous Random Variables
- ▶️ Continuous Uniform Distribution
- ▶️ Normal (Gaussian) Distribution

Next Class

- 🤩 Next class we will conclude our exploration of probability by looking into
  - ▶️ More Advanced Simulations!